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Department of Mathematics
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Math 208 Major Exam I Term 233

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. The solution space of the system

$$\begin{aligned} x_1 - 4x_2 - 3x_3 - 7x_4 &= 0 \\ 2x_1 - x_2 + x_3 + 7x_4 &= 0 \\ x_1 + 2x_2 + 3x_3 + 11x_4 &= 0 \end{aligned}$$

is the set of all linear combinations $s\mathbf{u} + t\mathbf{v}$ where

(a) $\mathbf{u} = (-1, 1, 1, 0)$, $\mathbf{v} = (-5, -3, 0, 1)$

(b) $\mathbf{u} = (1, -1, 1, 0)$, $\mathbf{v} = (-5, -3, 0, 1)$

(c) $\mathbf{u} = (-1, -1, 1, 0)$, $\mathbf{v} = (-5, 3, 0, 1)$

(d) $\mathbf{u} = (-1, -1, 1, 0)$, $\mathbf{v} = (-5, -3, 0, -1)$

(e) $\mathbf{u} = (-1, -1, 1, 0)$, $\mathbf{v} = (-5, -3, 0, 1)$

2. By making a suitable substitution, the differential equation $xyy' = x^2 + 3y^2$ can be transformed into a separable differential equation

(a) $\dfrac{v^2}{1+v^2}\,dv = \dfrac{1}{x}\,dx$

(b) $\dfrac{3v}{1+2v^2}\,dv = \dfrac{1}{x}\,dx$

(c) $\dfrac{v}{1+2v^2}\,dv = \dfrac{1}{x}\,dx$

(d) $\dfrac{2v}{1+2v^2}\,dv = \dfrac{1}{x}\,dx$

(e) $\dfrac{4v}{1+2v^2}\,dv = \dfrac{1}{x}\,dx$

3. A general solution of the differential equation $y'' = 2yy'$ is

(a) $y = A\tan(Ax + B)$

(b) $y = A\tan(Ax^2 + B)$

(c) $y = A\cot(Ax + B)$

(d) $y = A\tan(Ax^3 + B)$

(e) $y = A\cot(Ax^2 + B)$

4. For what values of k are the vectors $\mathbf{u} = (3, -4, 5)$, $\mathbf{v} = (1, k, 3)$, $\mathbf{w} = (0, 1, 4)$ of $\mathbb{R}^3$ linearly independent?

(a) $k \neq -2$

(b) $k \neq -1$

(c) $k \neq 0$

(d) $k \neq 3$

(e) $k \neq 2$

5. By making a suitable substitution, the differential equation $x\dfrac{dy}{dx} + y = y^{-2}$ can be transformed into a linear differential equation

(a) $\dfrac{du}{dx} + \dfrac{2}{x}u = \dfrac{3}{x}$

(b) $\dfrac{du}{dx} + \dfrac{3}{x}u = \dfrac{3}{x}$

(c) $\dfrac{du}{dx} - \dfrac{2}{x}u = \dfrac{3}{x}$

(d) $\dfrac{du}{dx} - \dfrac{3}{x}u = \dfrac{3}{x}$

(e) $\dfrac{du}{dx} - \dfrac{3}{x}u = \dfrac{1}{x}$

6. Find a constant k that makes the differential equation

$$(y^3 + kxy^4 - 2x)\,dx + (3xy^2 + 20x^2y^3)\,dy = 0$$

exact.

(a) 6

(b) -8

(c) -10

(d) 8

(e) 10

7. The general solution of the linear differential equation $xy' + (3x - 4)y = 6x^5$ is given by

(a) $y(x) = 2x^4 + cx^2e^{3x}$

(b) $y(x) = 2x^4 + cx^4e^{-3x}$

(c) $y(x) = 2x^2 + cx^2e^{3x}$

(d) $y(x) = 2x^4 + cx^3e^{-3x}$

(e) $y(x) = 2x^3 + cx^3e^{-3x}$

8. The sum of all values of r such that $y = x^r$ is a solution of the differential equation

$$x^2y'' - 2xy' - 4y = 0$$

is

(a) -4

(b) 0

(c) 4

(d) 3

(e) -3

9. The general solution of the separable differential equation $x^2\dfrac{dy}{dx} = y - xy$ is given by

(a) $\ln\left|\dfrac{x}{y}\right| - \dfrac{1}{x} = c$

(b) $2\ln\left|\dfrac{y}{x}\right| + \dfrac{1}{x} = c$

(c) $\ln|yx| + \dfrac{1}{x} = c$

(d) $\ln\left|\dfrac{y}{x}\right| + \dfrac{1}{x} = c$

(e) $2\ln|yx| + \dfrac{1}{x} = c$

10. The population of a town grows at a rate proportional to the population present at time t. The initial population was 500. After 10 years, the population is 575. The time needed for the population to reach 1000 is

(a) $\dfrac{10\ln 2}{\ln 23 - \ln 5}$

(b) $\dfrac{5\ln 2}{\ln 23 - \ln 10}$

(c) $\dfrac{2\ln 2}{\ln 23 - \ln 10}$

(d) $\dfrac{10\ln 2}{\ln 23 - \ln 20}$

(e) $\dfrac{20\ln 2}{\ln 23 - \ln 5}$

11. A general solution of the exact differential equation

$$(x - y^3 + y^2 \sin x)\, dx - (3xy^2 + 2y \cos x)\, dy = 0$$

is

(a) $xy^3 + y^2 \cos x - \frac{1}{2}x^2 = c$

(b) $xy^3 + y^2 \sin x - \frac{1}{2}x^2 = c$

(c) $xy^2 + y^3 \cos x - \frac{1}{2}x^2 = c$

(d) $xy^3 + y^2 \sin x - \frac{1}{2}x^3 = c$

(e) $xy^3 + y^2 \cos x + \frac{1}{2}x = c$

12. A general solution of the differential equation $\frac{dy}{dx} = 2 + \sqrt{y - 2x + 3}$ is

(a) $(x + c)^2 = 4(y + 2x - 3)$

(b) $(x + c)^2 = 4(y - 2x + 3)$

(c) $(x + c)^2 = 4(y - 2x - 3)$

(d) $(x + c)^3 = 4(y - 2x + 3)$

(e) $(x + c) = 4(y - 2x + 3)$

13. Which one of the following statements is true about the subset V of $\mathbb{R}^3$ defined by

$$V = \{(x_1, x_2, x_3) : \ x_3 \geq 0\}?$$

(a) V is closed under addition but not closed under multiplication by scalar

(b) V is not closed under addition and not closed under multiplication by scalar

(c) V is closed under multiplication by scalar

(d) V is a subspace of $\mathbb{R}^3$

(e) V is not closed under addition

14. Let $\mathbf{u} = (1, 2)$, $\mathbf{v} = (-1, 5)$, $\mathbf{w} = (8, -5)$ be vectors in $\mathbb{R}^2$. If $\mathbf{w} = a\mathbf{u} + b\mathbf{v}$, then $a + b =$

(a) 2

(b) 3

(c) 0

(d) 5

(e) 4

15. A particle is moving in a straight line with acceleration $a(t) = \dfrac{1}{\sqrt{t+9}}$, and initial position $x(0) = 4$, and an initial velocity $v(0) = 2$. Then $x(16)$ (the position of the particle at $t = 16$) is

(a) $\dfrac{214}{3}$

(b) $\dfrac{211}{3}$

(c) $\dfrac{209}{3}$

(d) $\dfrac{212}{3}$

(e) $\dfrac{217}{3}$

Answer Key

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Q	Ans	Brief explanation
1	**(e)**	$x_2 = -x_3 - 3x_4$, $x_1 = -x_3 - 5x_4$: $s(-1,-1,1,0) + t(-5,-3,0,1)$.
2	**(c)**	$y = vx$: $xvv' = 1 + 2v^2$, so $\frac{v}{1+2v^2}dv = \frac{1}{x}dx$.
3	**(a)**	$y' = y^2 + A^2$, so $\frac{1}{A}\arctan\frac{y}{A} = x + B_1$: $y = A\tan(Ax + B)$.
4	**(b)**	$\det = 12(k+1) \neq 0$, so $k \neq -1$.
5	**(b)**	Bernoulli $n = -2$, $u = y^3$: $u' + \frac{3}{x}u = \frac{3}{x}$.
6	**(e)**	$M_y = 3y^2 + 4kxy^3$, $N_x = 3y^2 + 40xy^3$, so $k = 10$.
7	**(b)**	$\rho = x^{-4}e^{3x}$, $(\rho y)' = 6e^{3x}$, so $y = 2x^4 + cx^4e^{-3x}$.
8	**(d)**	$r(r-1) - 2r - 4 = r^2 - 3r - 4 = 0$, roots 4 and -1, sum 3.
9	**(c)**	Separate: $\frac{dy}{y} = (\frac{1}{x^2} - \frac{1}{x})dx$, so $\ln\lvert yx\rvert + \frac{1}{x} = c$.
10	**(d)**	$P = 500e^{kt}$, $k = \frac{\ln 23 - \ln 20}{10}$, $t = \frac{\ln 2}{k} = \frac{10\ln 2}{\ln 23 - \ln 20}$.
11	**(a)**	$F = \frac{x^2}{2} - xy^3 - y^2\cos x$, so $xy^3 + y^2\cos x - \frac{1}{2}x^2 = c$.
12	**(b)**	$v = y - 2x + 3$: $v' = \sqrt{v}$, $2\sqrt{v} = x + c$, so $(x+c)^2 = 4(y - 2x + 3)$.
13	**(a)**	Sums keep $x_3 \geq 0$, but $(-1)(0,0,1) \notin V$: closed under addition only.
14	**(a)**	$a - b = 8$, $2a + 5b = -5$ give $a = 5$, $b = -3$, so $a + b = 2$.
15	**(d)**	$v = 2\sqrt{t+9} - 4$, $x = \frac{4}{3}(t+9)^{3/2} - 4t - 32$, so $x(16) = \frac{212}{3}$.